

Multiple Choice (20 marks)

1. In Python 3, which of the following function removes all leading and trailing whitespace in string?
 - (a) `replace(old, new [, max])`
 - (b) `strip([chars])`
 - (c) `swapcase()`
 - (d) `title()`
2. A large hospital maintains a list of patients' records in no particular order. To find the record of a given patient, which represents the most efficient...
 - (a) Do a sequential search on the name field of the records.
 - (b) Do a binary search on the name field of the records.
 - (c) Use insertion sort to sort the records alphabetically by name; then do a sequential search on the name field of the records.
 - (d) Use mergesort to sort the records alphabetically by name; then do a sequential search on the name field of the records.
3. In Python 3, `b = [11,13,15,17,19,21]; print(b[::2])` outputs what?
 - 19,21
 - 11,15
 - 11,15,19
 - 13,17,21
4. In Python 3, what is the following function returns item from the list with max value?
 - (a) `cmp(list)`
 - (b) `len(list)`
 - (c) `max(list)`
 - (d) `min(list)`
5. Let `l = [1,2,3,4]`. What is `sum(l)` in Python3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8

6. Which of the following school policies is most likely to have a positive impact on the digital divide?
- (a) A school allows students to bring a graphing calculator from home to complete in-class mathematics assignments.
 - (b) A school allows students to bring a tablet computer to class every day to participate in graded quizzes.
 - (c) A school provides a laptop or tablet computer to all students enrolled at the school.
 - (d) A school recommends that all students purchase a computer with as much processing speed as possible so that projects run faster.
7. In python 3, which of the following is floor division?
- (a) /
 - (b) //
 - (c) %
 - (d) —
8. A sorted list of numbers contains 500 elements. Which of the following is closest to the maximum number of list elements that will be examined when...
- (a) 10
 - (b) 50
 - (c) 250
 - (d) 500
9. In Python 3, which of the following function checks in a string that all characters are in uppercase?
- (a) isupper()
 - (b) join(seq)
 - (c) len(string)
 - (d) ljust(width[, fillchar])
10. In Python 3, which of the following function convert a string to an int in python?
- (a) int(x [,base])
 - (b) long(x [,base])
 - (c) float(x)
 - (d) str(x)

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Historically, it has been observed that computer processing speeds tend to double every two years. Which of the following best describes...' is 'Technology companies can accurately predict the dates when new computing innovations will be available to use.'
2. True or False: The correct answer to 'Consider the following numbers. ° Binary 1100 ° Decimal 11 ° Hexadecimal D Which of the following lists the numbers in order from least...' is 'Decimal 11, Binary 1100, Hexadecimal D'.
3. True or False: The correct answer to 'Which of the following best describes a Web server?' is 'A computer system running software that provides a user-friendly interface for creating Web pages'.
4. True or False: The correct answer to 'In Python 3, let $r = \lambda q: q * 2$. What is $r(3)$?' is '6'.
5. True or False: The correct answer to 'What is the role of the compiler in the process of developing executable software?' is 'Translating a program written in an abstract, high-level language into a program with the same behavior expressed in machine code'.
6. True or False: The correct answer to 'Let $x = 8$. What is $x \ll 1$ in Python 3?' is '8'.
7. True or False: The correct answer to 'Which of the following is the hexadecimal representation of the decimal number 231_{10} ?' is 'E4_{16}'.
8. True or False: The correct answer to 'Which of the following activities poses the greatest personal cybersecurity risk?' is 'Paying a bill using a secure electronic payment system'.
9. True or False: The correct answer to 'Which of the following has the greatest potential for compromising a user's personal privacy?' is 'A group of cookies stored by the user's Web browser'.
10. True or False: The correct answer to 'The code fragment below is intended to display "odd" if the positive number num is odd. IF (MISSING CONDITION) DISPLAY "odd" Which of...' is '(num MOD 1) = 1'.

End of Paper